

Name: solution (please print)

Signature: _____

ECE 2202 – Quiz 1

September 18, 2025

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.
2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.
3. You will have 30 minutes to work on the quiz. For online students: this time includes the time to print, scan, and upload your work. Points will be deducted if it is not uploaded by the deadline time.

_____/20

units (+2 pts)
organization (+1 pt)

Problem 1 (20 pts)

Using only KVL, KCL, and Ohm's Law (not Node Voltage Method) on the circuit below, find the following:

- Find v_Q .
- Find v_X .
- Find the power delivered by the current source.

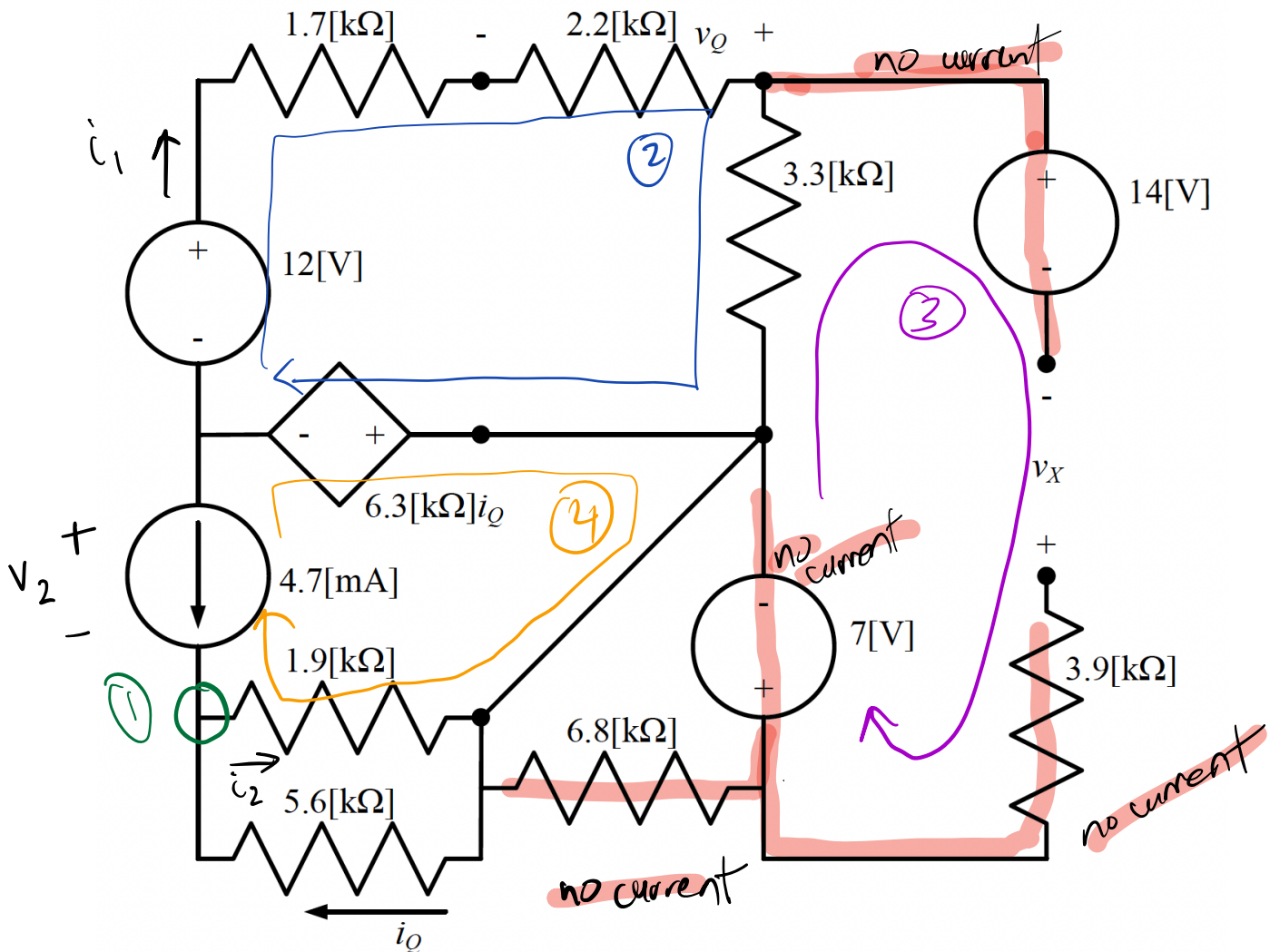


Figure 1

a) KCL ① $-4.7 [\text{mA}] + i_2 - i_Q = 0$ (+3)
 current divider: $-i_Q = 4.7 [\text{mA}] \left(\frac{1.9 [\text{k}\Omega]}{1.9 + 5.6 [\text{k}\Omega]} \right)$
 $i_Q = -1.19 [\text{mA}], i_2 = 3.51 [\text{mA}]$

$$\text{KVL } \textcircled{2} -12 [\text{V}] + i_1 \cdot 1.7 [\text{k}\Omega] + i_1 \cdot 2.2 [\text{k}\Omega] + i_1 \cdot 3.3 [\text{k}\Omega] + 6.3 [\text{k}\Omega] i_Q = 0 \quad \textcircled{+2}$$

$$i_1 = 2.71 [\text{mA}] \quad \textcircled{+1}$$

$$\textcircled{+3} v_Q = -i_1 \cdot 2.2 [\text{k}\Omega] = \boxed{-5.96 [\text{V}]} \quad \textcircled{+1}$$

$$\text{b) KVL } \textcircled{3} -i_1 \cdot 3.3 [\text{k}\Omega] + 14 [\text{V}] - v_X + 7 [\text{V}] = 0 \quad \textcircled{+3}$$

$$\boxed{v_X = 12.06 [\text{V}]} \quad \textcircled{+1}$$

$$\text{c) } P_{\text{DEL}, 4.7 [\text{mA}]} = -4.7 [\text{mA}] \cdot v_2 \quad \textcircled{+2}$$

$$\text{KVL } \textcircled{4} -v_2 - 6.3 [\text{k}\Omega] i_Q - i_2 \cdot 1.9 [\text{k}\Omega] = 0$$

$$v_2 = 0.83 [\text{V}]$$

$$\boxed{P_{\text{DEL}, 4.7 [\text{mA}]} = -3.89 [\text{mW}]} \quad \textcircled{+1}$$